

PRINCIPAL ARCHITECT & AI LEAD — INTERVIEW PRESENTATION

A Platform Built to Scale

Reasoning, trade-offs, and open questions

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Based on public information and prior conversations with Mark and Laurence. Current state assumptions to be validated on day one.

High Level Summary

THE PROPOSITION

- A smartphone camera measures blood pressure and pulse rate. Micro colour changes in facial skin (remote photoplethysmography) analysed by proprietary AI.
- CE Class IIa EU MDR certified for BP and pulse. Currently in pilot with NHS
- FDA clearance in progress.

WHERE THE VALUE SITS

- The science is proven with six patents secured.
- A data model of 25,000+ paired clinical recordings from 12,500 patients.
- Both systolic and diastolic to clinical grade.
- The value needs to be unlocked, getting this into the hands of 100 million people, not 100 thousand.
- The current solution doesn't scale, this is a platform problem.

THE TECHNICAL CHALLENGE

Parallel improvements required:

- Scale without breaking what's in production today
- Every change traceable for regulators
- Deliver with a small focused team

Top Five Areas

	T O D A Y	T A R G E T	A G E N T I C A I
Reduce cost	10p+ per assessment Cold container per request Economics break at scale	<1p per assessment Warm inference pool (AKS) On-device signal extraction	Cost thresholds monitored via Grafana alerting Ops agent correlates spikes with known change
Improve performance	No latency guarantee, users wait for model All compute in cloud Video transmitted from device	Async queue, warm pool Horizontal autoscaling Signal extracted on-device Metal/Vulkan GPU pipeline	Grafana alerting detects performance degradation ML Validation agent monitors output for model drift
Release safely at pace	Manual release gates No formal API contract Test evidence ad hoc	Automated pipeline Immutable artefacts Every sprint a release candidate	Compliance agent gates PRs Code Review agent on every PR Doc agent drafts changelogs
Observability	Limited visibility into production failures No mobile SNR telemetry No root cause analysis of failure	Grafana + Prometheus for metrics Loki for logs Sentry for error tracking Signal quality reported per assessment	Ops agent reads metrics Creates GitHub Issues from alerts AI surfaces, humans decide
Governance & planning	Decisions undocumented Sprint cadence informal Traceability manual	ADRs before code 1-week sprints, DoD with regulatory gates eQMS integrating with development process	Compliance agent enforces traceability at PR time Governance visible continuously

Reduce Cost

SUMMARY

Current architecture: A new container is created per assessment to guarantee patient data isolation, a safe pragmatic call, but expensive. Each cold start loads the ML model from scratch.

At low volumes this is manageable, it becomes a commercial problem as we scale.

TECH STACK

Current - Flask

Target - AKS · Azure Service Bus · FastAPI

APPROACH & RECOMMENDATION

Refactor to a stateless inference service enabling a warm pool where model weights stay loaded in memory. Patient isolation is guaranteed by design, not container boundary. Async queue (Azure Service Bus) absorbs demand spikes.

On-device signal extraction moves face detection, ROI extraction, and POS + CHROM processing to the device. Only the processed BVP signal is transmitted. Removes a further chunk of cloud compute per assessment.

ALTERNATIVES CONSIDERED & REJECTED

Serverless inference (Azure Functions) the cold start still happens, it's just hidden. Full stateless rewrite is the right destination but too risky to do in one go on a live clinical system. Incremental refactor is safer.

Improve Performance — Cloud & Client

SUMMARY

The phone is a signal processor, not just a camera. Performance improvements come from two places: better signal quality captured on-device, and faster inference in the cloud. Together they reduce total assessment time and failed assessments.

TECH STACK

Android - MediaPipe · Vulkan · Kotlin
iOS - Vision framework · Metal · Swift/SwiftUI

APPROACH & RECOMMENDATION

Client: Metal (iOS) and Vulkan (Android) GPU frameworks run the signal extraction pipeline on-device. MediaPipe tracks face landmarks continuously. ROI extraction targets forehead and cheeks. POS + CHROM fusion extracts the blood volume pulse signal. A signal quality gate rejects low-quality frames before transmission, only clean signal reaches the cloud.

Cloud: warm pool means inference begins the moment signal arrives, no cold start wait. Cleaner incoming signal means the model runs faster with higher confidence.

The ISP (image signal processor) has auto-exposure, auto-white-balance, H.264 compression which degrades the rPPG signal. The on-device pipeline is designed to work around this.

ALTERNATIVES CONSIDERED & REJECTED

All compute in cloud, raw video transmitted, simpler day one but slower, phase 1 reality while stateless refactor is in progress. Shared C++ core across iOS and Android, reduces duplication but sacrifices platform-native performance.

Release Safely at Pace

SUMMARY

Release controls that don't slow the process. Everyone knows what is happening and consistency drives the right behaviour. In a regulated medical device the control is not optional, it is the evidence regulators need to trust what you ship.

TECH STACK

GitHub Actions · ACR · TestRail · SonarCloud
· Snyk · Syft · Qualio

APPROACH & RECOMMENDATION

Pipeline gates in order: build and unit tests, SonarCloud static analysis, Snyk dependency scan, AI code review, Syft SBOM generation, TestRail trigger, human approval to production. No automated promotion to production.

Three-layer test strategy: unit and integration tests automated via GitHub Actions. System verification in TestRail, requirement to test case to execution to result, this is the IEC 62304 regulatory artefact. Clinical performance validation to FDA submission standard, separate from software verification.

Every sprint produces a release candidate. Production requires PO sign-off, QA gate, and regulatory checklist.

ALTERNATIVES CONSIDERED & REJECTED

Informal test evidence back-filled before submission. The cost of back-filling is always higher than the cost of discipline. Rejected. Fully automated promotion to production. Not compatible with regulated software sign-off requirements. Rejected. Compliance tooling separate from the delivery pipeline. Creates two systems of record, engineers treat compliance as something that happens to them rather than something they own. Rejected.

Trade-off: compliance gates add friction. The compliance agent makes traceability linkage visible at PR time — engineers see the gap before it becomes a submission problem, not after. AI code review without measuring action rate risks becoming noise engineers route around.

Observability

SUMMARY

If something goes wrong in the production you need to know before a clinician does. A failed assessment is not just a UX problem, it is a trust problem. You cannot improve what you cannot see.

TECH STACK

Grafana · Prometheus · Loki · Sentry · PagerDuty · Sentry Mobile SDK

APPROACH & RECOMMENDATION

Cloud: Grafana and Prometheus for infrastructure and service metrics. Loki for structured log aggregation. Sentry for error tracking correlated to specific releases. PagerDuty for on-call routing. Key metrics: assessment success rate, inference latency p50/p95/p99, queue depth, cost per assessment.

Mobile: signal quality telemetry returned post assessment covering SNR per ROI, motion artefact rate, rejection rate, and confidence scores. Sentry mobile SDK for crash rate and B2B integration errors.

Privacy constraint: telemetry is aggregated numbers. Raw video never leaves the device.

ALTERNATIVES CONSIDERED & REJECTED

Azure Monitor only, less flexible for correlating mobile and cloud signals in a single view. No mobile telemetry in phase 1, rejected, without signal quality data from production you are optimising blind. Anecdotal AI ROI measurement is not acceptable in a regulated environment where tooling decisions need evidence.

Governance, Planning & Control

SUMMARY

Be explicit about what good looks like, then follow through. If you keep risk logs, decision records, and documentation as part of everyday work, an audit is just someone reading what you already wrote.

TECH STACK

GitHub Projects · Qualio · ADR format in repo · Slack

APPROACH & RECOMMENDATION

ADRs written before implementation containing; context, decision, rationale, alternatives rejected. Stored in the repo, reviewed via PR. Evidences why decision were made.

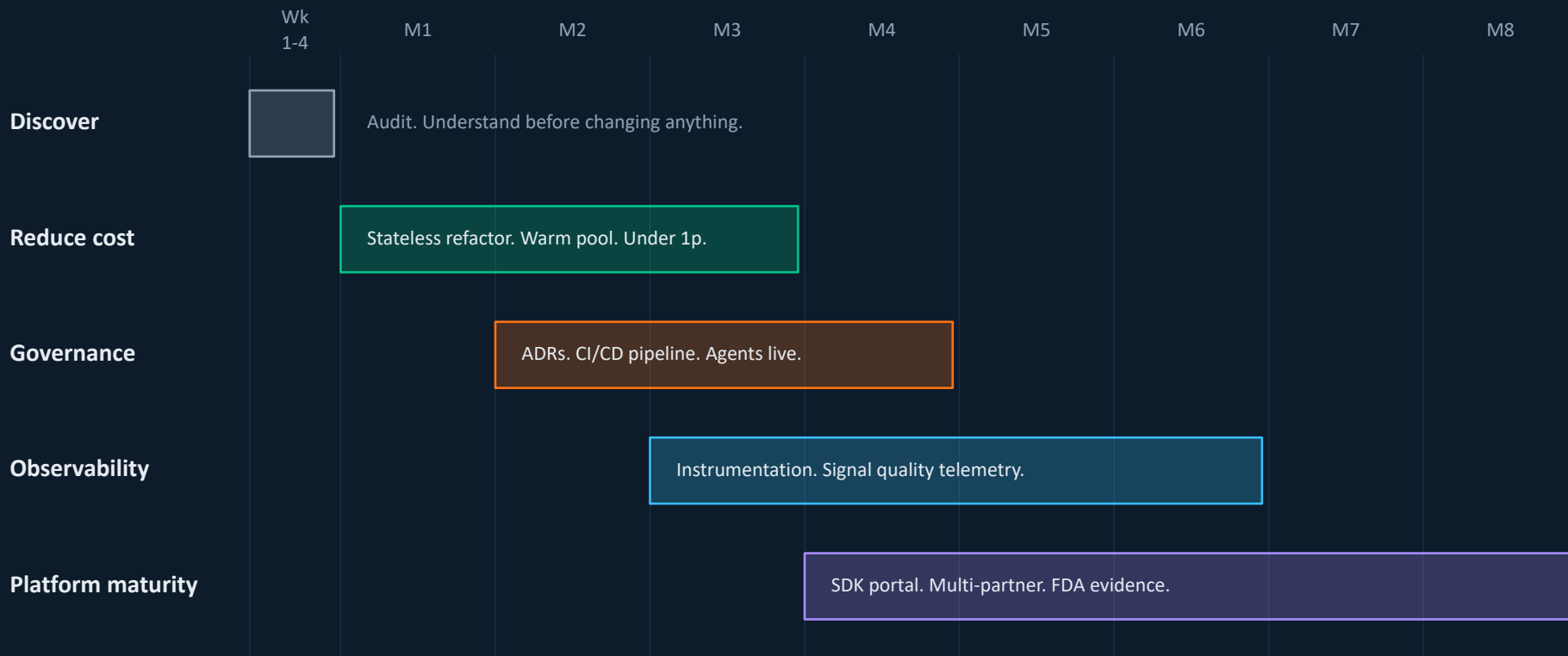
Definition of Done includes regulatory gates. Risk class assigned, traceability link present, TestRail execution complete, PO acceptance. Not optional, not completed retrospectively.

One week sprints, planning driven by value and risk. Mapping what is genuinely differentiating against what is just the cost of playing in this space.

ALTERNATIVES CONSIDERED & REJECTED

Formal architecture review board, rejected, ADRs via PR achieve the same outcome with a fraction of the overhead. Velocity only planning, rejected, planning must reflect regulatory risk, a story touching a clinically significant component has a different cost to one that does not. Governance tooling separate from the delivery pipeline, rejected, if compliance is not in the workflow engineers treat it as something that happens to them rather than something they own.

Incremental improvements without impacting production



Phases overlap. Sequencing reflects priority not exclusivity. MVP of each phase is brought forward where needed — e.g. instrumentation during the stateless refactor, not after.

Appendix — Platform Technology Stack



Current



Phase 1



Phase 2



Phase 3+

Cloud Infrastructure

Azure Container Apps	Container hosting, warm pool
AKS	Container orchestration at scale
Azure Service Bus	Async assessment queue
Azure Cache for Redis	Stateless session state
ACR	Immutable artefact registry
Terraform	Infrastructure as Code

Backend & Inference

Python / Flask	Current inference service
FastAPI	Target async inference service
Triton Inference Server	ML model serving at scale
MLflow	Experiment tracking, model registry

Signal Processing

POS + CHROM algorithms	rPPG signal extraction
SNR-weighted fusion	Best signal quality combination
Signal quality gate	Reject low-SNR before transmission

Mobile — iOS

Swift / SwiftUI	iOS SDK language and UI
Apple Vision Framework	Face landmark detection (native)
Metal	GPU compute for signal extraction
Fastlane	Build automation and release

Mobile — Android

Kotlin	Android SDK language
MediaPipe	Cross-platform face landmark tracking
Vulkan	GPU compute for signal extraction

Testing & Verification

pytest	Unit and integration tests
TestRail	Test management, IEC 62304 §5.7 artefact
Renovate / Dependabot	Automated dependency updates
Bland-Altman analysis	Clinical performance validation

Developer Experience

GitHub Pages → Backstage	Docs and internal developer portal
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CI/CD Pipeline

GitHub Actions	Pipeline execution, immutable logs
SonarCloud	Static analysis quality gate
Snyk	Dependency and container vulnerability scan
Syft	SBOM generation (FDA cybersecurity artefact)
Trivy	Container image scanning
AI Code Review Agent	Structured PR review

Observability — Cloud

Grafana	Metrics dashboards and alerting
Prometheus	Infrastructure and service metrics
Loki	Structured log aggregation
Sentry	Error tracking, release correlation
PagerDuty	On-call routing and incident management
Azure Monitor	Native monitoring (considered)

Observability — Mobile

Sentry Mobile SDK	Crash rate, ANR, B2B integration errors
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Appendix — Governance, Security, AI Agents & Regulatory Standards

Governance & QMS

GitHub / GitHub Projects

Source control and sprint planning

Qualio

eQMS — controlled docs, CAPA, training records

ADR format (in repo)

Architecture Decision Records

Slack

Communication only — not a compliance record

Wardley / Value mapping

Value and risk-driven planning

Security

Snyk

Dependency and container vulnerability scanning

Syft

SBOM generation every build

Trivy

Container image scanning

Azure Key Vault

Secrets and certificate management

Cyber Essentials Plus

UK government security baseline

AI Agents

Anthropic Claude API

Primary agentic framework

Code Review Agent

Structured PR review (Z2)

Compliance Agent

Traceability gate at PR time (Z2)

Documentation Agent

Merge-triggered changelog (Z2)

Ops Agent

Alert-triggered issue creation (Z3)

ML Validation Agent

Model drift detection (Z2)

Z2 = Review & CI Gate · Z3 = Operations

Regulatory Standards

IEC 62304

Medical device software lifecycle

ISO 13485

Quality management system

ISO 14971

Risk management

EU MDR Class IIa

CE certification framework

FDA 510(k) / De Novo

US regulatory pathway

GDPR

Data protection

NHS DSPT

NHS data security toolkit

AAMI/ESH/ISO BP standard

Clinical accuracy threshold ($\leq 5 \pm 8$ mmHg)

FDA Cuffless BP Guidance

Jan 2026 clinical performance testing

FDA Cybersecurity Guidance

2023 — SBOM and vulnerability mgmt

Note on SBOM

This SBOM reflects the target platform architecture as described in the presentation. Current state to be validated on day one. In the target architecture, Syft generates a formal machine-readable SBOM (CycloneDX or SPDX format) on every build as a required FDA cybersecurity artefact.